PAWS Storefront Design

Carlos Cain

CST-452 Capstone Implementation Plan

Grand Canyon University

Instructor: Professor Amr Elchouemi

Revision: v1.1

Date: February 15, 2026

© 2025. Grand Canyon University. All Rights Reserved.

PAWS Storefront – Full Scrum Implementation Plan

1. Product Backlog (User Stories)

These come directly from your design document's flows, routes, and use cases.

User Story ID	User Story Description
US-01	As a user, I want to register an account so I can log in securely.
US-02	As a user, I want to log in so I can access protected features.
US-03	As a user, I want to browse products so I can view available items.
US-04	As a user, I want to add items to my cart so I can purchase them.
US-05	As a user, I want to view my cart so I can see selected items.
US-06	As a user, I want to remove items from my cart so I can adjust my order.
US-07	As a user, I want to check out so I can complete my purchase.
US-08	As a user, I want to see an order confirmation, so I know my purchase succeeded.
US-09	As a user, I want to view my order history so I can track past purchases.
US-10	As an admin, I want to create products so I can manage inventory.
US-11	As an admin, I want to update products so I can correct or adjust listings.
US-12	As an admin, I want to delete products so I can remove outdated items.
US-13	As the system, I must enforce role-based access control for security.

© 2025. Grand Canyon University. All Rights Reserved.

2. Sprint Breakdown (4 Sprints)

Sprint 1 : Foundation & Authentication (Week 1)

Goal: Set up project, database, security, registration, login.

Task	Est. Hours	Actual Hours	Status
Project setup (Spring Boot, packages, configs)	4	4	Complete
Create database schema (Users, Authorities)	3	3	Complete
Implement RegistrationController + RegisterModel	4	5	Complete
Implement UserService + password hashing	3	3	Complete
Implement LoginController + LoginModel	3	3	Complete
Configure Spring Security (roles, login, logout)	5	6	Complete
Build registration.html + login.html	4	4	Complete

Sprint 1 Total:

Estimated: 26 hrs

Actual: 28 hrs

Completion: 100%

© 2025. Grand Canyon University. All Rights Reserved.

Sprint 2: Products & Cart (Week 2)

Goal: Product listing, product CRUD, cart functionality.

Task	Est. Hours	Actual Hours	Status
Implement ProductDao + ProductService	4	4	Complete
Implement ProductController	4	5	Complete
Build products.html	3	3	Complete
Implement CartItemDao + CartService	4	4	Complete
Implement CartController (add/view/remove)	5	6	Complete
Build cart.html	3	3	Complete
Admin: createProduct.html + updateProduct.html	4	4	Complete

Sprint 2 Total:

Estimated: 27 hrs

Actual: 29 hrs

Completion: 100%

© 2025. Grand Canyon University. All Rights Reserved.

Sprint 3: Checkout, Orders, Order History (Week 3)

Goal: Checkout flow, order creation, order details, confirmation page.

Task	Est. Hours	Actual Hours	Status
Implement OrderDao + OrderDetailDao	5	5	Complete
Implement OrderService (create order, details, totals)	5	6	Complete
Implement CheckoutController	4	4	Complete
Build checkout.html	3	3	Complete
Implement ConfirmationController	2	2	Complete
Build confirmation.html	2	2	Complete
Implement OrderHistoryController	3	3	Complete
Build orderHistory.html	3	3	Complete

Sprint 3 Total:

Estimated: 27 hrs

Actual: 28 hrs

Completion: 100%

© 2025. Grand Canyon University. All Rights Reserved.

Sprint 4: Testing and Documentation (Week 4 - 6)

Goal: Test cases, bug fixes, UI polish, documentation, demo prep.

Task	Est. Hours	Actual Hours	Status
Write 27 test cases (TC-001 → TC-027)	6	7	Complete
Execute all test cases	4	4	Complete
Fix cart/userId issues	3	3	Complete
UI polish (Bootstrap responsiveness)	3	3	Complete
Finalize Source Code Listing	3	3	Complete
Finalize Traceability Matrix	2	2	Complete
Security Upgrade (TLS)	5	7	Complete

Sprint 4 Total:

Estimated: 26 hrs

Actual: 29 hrs

Completion: 100%

© 2025. Grand Canyon University. All Rights Reserved.

3. Total Project Hours

Category	Estimated	Actual
Sprint 1	26	28
Sprint 2	27	29
Sprint 3	27	28
Sprint 4	26	29
Total	**106 hrs**	**114 hrs**

© 2025. Grand Canyon University. All Rights Reserved.

4. Percent Complete

Sprint	Percent Complete
Sprint 1	100%
Sprint 2	100%
Sprint 3	100%
Sprint 4	100%
Overall Project Completion	**100%**

© 2025. Grand Canyon University. All Rights Reserved.

5. Burndown Summary (Scrum)

- Sprint 1: Authentication foundation complete
- Sprint 2: Product + Cart complete
- Sprint 3: Checkout + Orders complete
- Sprint 4: Testing + Documentation complete

Velocity: **~27 hours per sprint**

© 2025. Grand Canyon University. All Rights Reserved.